

Mostly Real, Locally Manipulated: CLAIMFORGE, a Paired Mouse-Core Stress Test for Localized AI Edits

Anonymous submission

Abstract

A small object insertion can change the apparent meaning of a consumer-evidence photograph even when nearly every pixel still comes from a real camera image. We introduce the canonical mouse core of CLAIMFORGE, a paired stress test for small AI object insertions in restaurant and lodging photographs. The core contains 275 human-approved source/forged pairs. Its pre-encoding exact RGB-difference masks cover a median 0.11264% of the image, after which both sides of every pair are freshly encoded with identical JPEG settings and inherited EXIF stripped. We reproduce eight open-source local-forensics methods under their registered native detection and localization capabilities. TruFor is the strongest joint research method, reaching 0.8179 image AUROC, 0.4436 TPR at 5% FPR, and 0.5001 macro pixel F1. CAT-Net reaches the highest pixel AP, 0.6123, but has no registered image-level head. On this canonical condition, no reproduced method simultaneously reaches the study-specific base-condition references of 0.90 AUROC and 0.50 macro pixel F1. Performance also varies sharply with edit size: TruFor AUROC rises from 0.7223 in the smallest quintile to 0.9412 in the largest. These results reveal meaningful forensic signal, but not reliable coverage of both capabilities motivated by consumer-claim review.

1 Introduction

Consumer services increasingly use photographs as evidence: a guest reports a pest in a hotel room, a diner reports contamination in a meal, or a tenant reports damage in a rental. Such images can affect refunds, platform penalties, reputation, and insurance decisions, yet they are captured in uncontrolled settings and often reviewed under time pressure. Modern editing changes the threat. An attacker need not synthesize a whole scene; a small, plausible object inserted into a real photograph can change the meaning of the claim.

This paper studies that mostly-real threat through the current canonical mouse core of CLAIMFORGE. Each forged image is paired with its real source, and the pre-encoding exact RGB differences occupy a median 0.11264% of the frame. Real and forged inputs receive the same final JPEG encoding. The setting is therefore different from whole-image

generation: global statistics are dominated by the source photograph, while a useful review tool should provide both a reliable image-level warning and spatial evidence identifying what changed.

We ask three questions. **RQ1:** Can current methods detect these tiny edits at low false-positive rates? **RQ2:** Can they localize the changed evidence accurately enough for review? **RQ3:** How does the strongest joint model vary with edit size and descriptive domain slices? We answer these questions on 275 human-approved real/forged pairs using eight fully reproduced research methods. We preserve methods' registered capabilities: a localization-only method remains N/A for detection rather than receiving an improvised score.

We make four evidence-backed contributions:

- We formalize a consumer-evidence threat and paired tasks for image-level manipulation detection and exact changed-pixel localization.
- We construct and audit a canonical mouse core with human quality control, matched container encoding, and exact-difference masks.
- We introduce capability-aware reporting that preserves N/A outputs and evaluates low-FPR detection, localization, and pristine-image false positives.
- We show a substantive but incomplete signal: TruFor is the strongest joint model, CAT-Net is a strong localization-only model, and no reproduced method clears both study-specific base-condition references on the canonical mouse condition.

The conclusion is deliberately scoped. We do not claim cross-editor or laundering robustness, and we do not transfer mouse-core results to the unevaluated cat and trash-can expansions. The current result is a paired stress test of one difficult, operationally relevant condition.

Evaluation Gap

Whole-image detection. GenImage evaluates image-level detection over complete synthetic images and multiple generators (Zhu et al. 2023). DIRE uses diffusion reconstruction discrepancies (Wang et al. 2023), while CLIP-based variants target cross-category generalization (Tan et al. 2024). These methods matter operationally, but a whole-image score neither guarantees sensitivity to a tiny insertion nor identifies the evidence an adjudicator should inspect.

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Manipulation localization and inpainting. TruFor combines an integrity score, localization, and reliability maps (Guillaro et al. 2022); HiFi-IFDL studies fine-grained detection and localization (Guo et al. 2023); and IMDL-BenCo unifies models and evaluation protocols (Ma et al. 2024). GIM, OpenSDI, and COCO-Inpaint move toward diffusion inpainting with both image- and pixel-level evaluation (Chen et al. 2024; Wang, Huang, and Hong 2025; Yan et al. 2025). CLAIMFORGE differs in the conjunction of a consumer-evidence threat, extremely small semantic insertions, source-paired inputs, and capability-aware reporting.

Shortcuts and missing evaluation. “Fake or JPEG?” shows that format differences can dominate generated-image benchmarks (Grommelt et al. 2024), while calibration work warns against interpreting a single fixed threshold as separability (Yang et al. 2026). Inpainting Exchange shows that global-artifact detectors can degrade when local editing removes those traces (Nebioglu, Bilgiç, and Popescu 2026). Mesorch and SAFIRE represent newer localization approaches (Zhu et al. 2024; Kwon et al. 2024). The unresolved evaluation gap is not simply another inpainting dataset: it is whether methods expose both low-FPR detection and actionable localization when nearly the whole consumer-evidence image is real.

2 CLAIMFORGE Mouse-Core Construction Threat and Tasks

Each example starts from a real source image x and a human-marked context and insertion region. An image editor produces a modified context crop; a difference-derived object mask controls paste-back into the source frame to produce $\tilde{x}$, and the final composite is manually reviewed. The target contains an image label $y \in \{0, 1\}$ and a pre-encoding exact RGB-difference mask M .

The stress test has two required tasks:

T1: Detection. Predict whether an image is real or locally manipulated.

T2: Localization. Predict the pixels changed by the local manipulation.

The consumer-claim threat model motivates evaluating both: an image-level alert without spatial support and a mask without a reliable alert each leave a distinct review gap.

Source Screening, Generation, and Human QA

The source pool contains 600 screened restaurant and lodging candidates. The annotation manifests contain 594 mouse-generation slots; six screened candidates do not appear as annotated slots, and their exclusion reasons must be reconciled in the final release documentation. Human annotators marked one context and insertion region for each slot. The recorded generation workflow produced edited context crops, and difference-derived object-only masks controlled paste-back before review of the final composites. The 147 lodging records identify HunyuanImage-3, while the 128 restaurant records retain the legacy editor tag `new_test`; exact editor provenance for those restaurant pairs is a release-blocking reconciliation item. Of 594 attempts, 275 were accepted, 288

Data stage	Count
Screened source candidates	600
Reviewed mouse attempts	594
Accepted mouse forgeries	275
Canonical real/forged pairs	275
Canonical images	550
Lodging / restaurant pairs	147 / 128
Selected cat expansion	251
Selected trash-can expansion	250

Table 1: Data-accounting funnel. Only the 275 canonical mouse pairs enter the results; the six-candidate screening/annotation gap still requires manifest reconciliation, and the cat/trash-can selections are unevaluated expansion assets.

were rejected for edit quality, and 31 had unsuitable sources. The accepted core contains 147 lodging and 128 restaurant pairs.

Canonical Mouse Core

The paired release independently decodes the source and reviewed forged PNG to RGB, then freshly encodes both as JPEG at quality 95 with 4:4:4 subsampling and inherited EXIF stripped. The localization target is computed before this final encoding as the exact nonzero RGB difference. The builder rejects an empty target, a size mismatch, an invalid box, or any changed pixel outside the reviewed context. It records content hashes for inputs, masks, and manifests.

Exact changed pixels occupy 0.02486%–1.29272% of an image, with median 0.11264% and mean 0.16846%. The 275 registered task pairs contain 270 unique pristine image hashes because five sources appear twice. The current confidence intervals resample task pairs; Section 6 discusses the resulting dependence.

Scope Boundary

The repository also contains 251 selected cat and 250 selected trash-can variants. They are not part of the common detector matrix and are excluded from all headline results. Their cross-decoder pixel-integrity audit must be resolved before a broader multi-object release or claim. Likewise, second-editor, laundering, real-object paste-back, genuine-object, and human studies remain future evaluation rather than completed controls.

3 Capability-Aware Evaluation Protocol Reproduced Methods and Native Outputs

The primary comparison reproduces MaskCLIP (Wang, Huang, and Hong 2025), TruFor (Guillaro et al. 2022), CAT-Net v2, MVSS-Net, PSCC-Net, and IML-ViT through their registered releases or the IMDL-BenCo harness (Ma et al. 2024), plus HiFi-IFDL (Guo et al. 2023) and Mesorch (Zhu et al. 2024). We use pinned public checkpoints and registered preprocessing, and report only capabilities exposed by the registered method. A localization-only method receives T1=N/A; we do not convert map means or maxima into an

Method	Registered T1	AUROC	TPR@5%	Pixel AP	Macro F1	Macro IoU	Real FP area
TruFor	yes	0.8179	0.4436	0.5799	0.5001	0.3741	1.776%
CAT-Net v2	no	N/A	N/A	0.6123	0.4748	0.3574	0.436%
IML-ViT	no	N/A	N/A	0.1551	0.1433	0.0984	0.802%
Mesorch	no	N/A	N/A	0.0922	0.0454	0.0314	0.358%
MaskCLIP	yes	0.5073	0.0509	0.0474	0.0056	0.0037	0.060%
MVSS-Net	yes	0.5058	0.0473	0.0341	0.0196	0.0125	3.289%
PSCC-Net	yes	0.5041	0.0473	0.0153	< 0.001	< 0.001	0.827%
HiFi-IFDL	yes	0.4992	0.0509	0.0036	< 0.001	< 0.001	0.729%

Table 2: Primary zero-shot results on 275 canonical real/forged mouse pairs. TPR@5% is measured at empirical FPR no greater than 5%. Pixel metrics are per-image macro values on forged images at each registered threshold; pixel AP uses the continuous map. Real FP area is the mean thresholded positive fraction on pristine images. A missing registered image-level output is reported as N/A rather than manufactured from a map.

unofficial detector. This preserves the distinction between a method that fails a task and one that does not implement it.

Metrics

For T1 we compute AUROC, average precision, TPR at empirical FPR no greater than 5%, the registered fixed-threshold decision, and paired score changes. Threshold-independent metrics are necessary because calibration can hide separability (Yang et al. 2026). For T2 we compute pixel AP, macro and micro F1/IoU/MCC at the public threshold, edit-box hits, and positive area on pristine images. Table 2 summarizes the decision-relevant core; the full metric matrix and uncertainty are assigned to the supplement. We treat detection and localization as two capability dimensions rather than collapsing them into a scalar.

Reference Target and Statistics

The experiment design specifies base-condition references of image AUROC at least 0.90 and macro pixel F1 at least 0.50. Its broader proposed “solved” rule also requires retaining both thresholds after JPEG-75 laundering, which has not yet been evaluated. We therefore use the two thresholds only as study-specific comparison points and make no claim under the full composite rule. Existing confidence intervals use 1,000 paired-task bootstrap replicates. Because five pristine sources repeat, these intervals are provisional and may be slightly optimistic; a source-hash-cluster bootstrap is required before submission.

Separate Diagnostic Protocols

The available four-run multimodal export is a legacy diagnostic: real inputs are source JPEGs while forged inputs are PNGs, and its boxes are not comparable with exact-difference pixel masks. It cannot support a model-performance claim and must be rerun on matched canonical JPEGs. Commercial services use heterogeneous inputs, coverage, score semantics, and mostly forged-only runs. We therefore report these systems only as diagnostics with explicit denominators and never rank their vendor-threshold rates against the primary paired AUROC table.

4 Main Results on the Canonical Mouse Core

Image-Level Detection

TruFor provides the strongest image-level signal: AUROC 0.8179 (provisional paired-task bootstrap 95% CI 0.7902–0.8447), average precision 0.8393, and TPR 0.4436 at FPR no greater than 5%. At its public 0.5 threshold it still misses 140/275 forged images. MaskCLIP, MVSS-Net, PSCC-Net, and HiFi-IFDL remain near 0.5 AUROC. CAT-Net, IML-ViT, and Mesorch expose no registered T1 head and remain N/A.

Pixel-Level Localization

CAT-Net has the best continuous pixel ranking, with macro pixel AP 0.6123. TruFor has the best registered-threshold macro F1 point estimate, 0.5001, and macro IoU, 0.3741. Its provisional paired-task bootstrap F1 interval, 0.4676–0.5318, crosses the 0.50 reference. This strength is not sufficient by itself: at the same threshold, 273/275 pristine images contain at least one TruFor-positive pixel, and their mean positive area is 1.776%. Mask quality must therefore be interpreted together with an image-level decision.

No Method Covers Both Capabilities

Among the eight reproduced methods evaluated under their registered native capabilities, none simultaneously reaches the base-condition references of AUROC 0.90 and macro pixel F1 0.50 on the canonical mouse core. TruFor’s localization point estimate narrowly reaches 0.50, although its provisional interval crosses that reference, while its AUROC remains below 0.90; CAT-Net is a strong localizer without a registered detector. This result demonstrates a capability gap rather than universal random behavior.

The reported confidence intervals currently resample task pairs. Five pristine image hashes occur twice among the 275 registered pairs, so pair-bootstrap intervals can be slightly optimistic. We retain the frozen protocol for comparability across methods and identify source-cluster bootstrap as a required final analysis.

5 Stress Patterns and Diagnostic Comparisons

Edit Size and Domain

In the available TruFor slices, edit size is the strongest observed pattern. TruFor AUROC rises from 0.7223 in the smallest exact-edit quintile to 0.9412 in the largest; macro pixel AP rises from 0.4240 to 0.7393. Lodging is descriptively easier than restaurant for TruFor (AUROC 0.8416 versus 0.7913), but this slice is not designed for a causal domain comparison.

Conditional Versus End-to-End Localization

Commercial outputs illustrate why localization quality must include detection misses. In a forged-only run, Copyleaks returns a native mask for 111/275 mouse forgeries; on those positive cases the mean IoU is 0.8165, but treating 164 empty-mask misses as failures reduces all-image mean IoU to 0.3296. The run cannot estimate real-image false-positive rate, so it demonstrates conditional localization rather than reliable end-to-end coverage.

Separate Diagnostic Evidence

The legacy four-run MLLM export uses source-JPEG real images and PNG forgeries, and its localization protocol returns boxes rather than exact changed-pixel masks. We retain it only as a protocol-warning artifact; any MLLM performance claim awaits a matched-JPEG rerun. Commercial services are likewise reported by coverage and vendor-threshold verdicts because most runs are forged-only and cannot estimate paired AUROC or false-positive rate.

Operational Takeaways

Our results motivate evaluation protocols that jointly report low-FPR image detection, spatial evidence, explicit coverage, and defined empty-mask behavior. A whole-image score alone does not establish sensitivity to local AI edits, while an attractive mask on selected positives can hide a high miss rate.

6 Limitations, Ethics, and Conclusion

Limitations

The common evaluation uses one recorded generation workflow and one fully evaluated object class; exact editor provenance for the 128 restaurant pairs requires manifest reconciliation. It does not yet test laundering, a held-out editor, whole-image contrast sets, real-object paste-back, genuine-object controls, or humans. Five pristine sources repeat, and source-cluster uncertainty remains to be computed. The selected cat and trash-can expansions are excluded pending common canonicalization and resolution of a cross-decoder pixel-integrity audit. The legacy MLLM export is format-confounded, and commercial results are heterogeneous diagnostics rather than a balanced leaderboard. Accordingly, our conclusion is limited to the canonical mouse condition.

Ethics and Release

Plausible manipulated evidence is dual-use. CLAIMFORGE is intended to help platforms, insurers, and forensic researchers evaluate defenses, but unrestricted release could also lower the cost of misuse. We therefore plan a gated, research-use release with source-license documentation, provenance, and explicit restrictions on operational fraud. This policy remains a release plan until implemented.

Conclusion

Relative to the decoded pre-canonical source, a median manipulated pair remains 99.887% unchanged while its apparent claim-relevant meaning changes. On 275 paired mouse edits, current methods expose useful but incomplete forensic signal: the strongest joint model misses the image-level reference, and the strongest continuous localizer lacks a registered detector. The immediate next steps are source-cluster uncertainty, provenance reconciliation, audited multi-object canonicalization, and realistic laundering tests.

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